Exercise 1: Recap of Knowledge Graphs

Task 1: Knowledge Graph Creation

Given are the following prefixes, namespaces, vocabulary terms and instances from DBpedia.

Prefixes and Namespaces	Vocabulary Terms	Instances

`dbo: https://dbpedia.org/ontology/`
 	
`dbr: https://dbpedia.org/resource/`
 	
`owl: http://www.w3.org/2002/07/owl#`
 	
`rdf:`
`http://www.w3.org/1999/02/22-rdf-syntax-ns#`
 	
`rdfs: http://www.w3.org/2000/01/rdf-schema#`
 	
`xsd: http://www.w3.org/2001/XMLSchema#`
	Properties:
	`dbo:capital`
	`dbo:country`
	`dbo:populationTotal`
	`rdf:type`
	`rdfs:label`
	
	Classes:
	`dbo:City`
	`dbo:Country`
	
	Data types:
	`xsd:nonNegativeInteger`
		`dbr:Berlin`
		`dbr:Germany`

Task 1.1 Draw a knowledge graph representing Berlin as the capital of Germany that has 3.645 million inhabitants using the vocabulary provided. Provide German labels of Germany and Berlin.

Task 1.2 Create the set of triples representing the knowledge graph in Task 1.1 using the Turtle syntax[1]. Hint: You can verify your .ttl file using an RDF Validator (https://www.w3.org/2015/03/ShExValidata/). With the RDF validator, check under "Status Summary" that "Schema" and "Data" are confirmed.

Task 1.3 Assume you want to model that Berlin has 564 bridges. What would you need to do? Write down the triples you need to model this.

Task 2: Queries on the DBpedia SPARQL endpoint

Task 2.1 Compare your results of Task 1.2 with the DBpedia page about Berlin (https://dbpedia.org/resource/Berlin). Can you re-identify the triples?

Task 2.2 Explore the DBpedia pages about `dbo:City`, `dbo:capital` and `dbo:populationTotal`. What are important properties to describe (i) classes, (ii) properties, (iii) relationships between classes/properties and (iv) the data types to be used on specific properties?

[1] https://www.w3.org/TR/turtle/

Task 2.3 What do you expect the following SPARQL query to return?

```
SELECT ?inhabitants WHERE {
      dbr:Germany dbo:capital ?capital .
      ?capital dbo:populationTotal ?inhabitants .
}
```

Task 2.4 Execute the query from Task 2.3 on DBpedia's SPARQL endpoint[2]. What is the output?

Task 2.5 Based on the query in Task 2.3, write a SPARQL query that returns all capitals of countries in the world and their inhabitants and execute it on the DBpedia SPARQL endpoint.

Task 3: Knowledge Graph Loading and Querying using RDFLib

➢ Jupyter notebook *Exercise1.ipynb*, Task 3.

Task 4: Knowledge Graph Exploration

Task 4.1 Explore different representations of Berlin in the following online knowledge graphs. What can you observe?

- Wikidata[3]
- DBpedia[4]
- Open Research Knowledge Graph[5]

Task 4.2 Which links between different representations of Berlin in different knowledge graphs exist?

Task 4.3 If no link between the same real-world entities in two knowledge graphs yet, how could you detect them?

[2] https://dbpedia.org/sparql
[3] https://www.wikidata.org/
[4] https://dbpedia.org/page/Berlin
[5] https://www.orkg.org/orkg/